MGT5493 – Programming in Business Analytics

Lab 1: NumPy & pandas Fundamentals for Business Data

Duration: 2 hours Dataset: store_sales.csv Tools: Jupyter Notebook

Lab 1 Objectives

1. Load and inspect a dataset in pandas.
2. Summarize data using descriptive statistics and simple aggregations.
3. Identify performance patterns across business units.
4. Create and interpret a basic visual summary of results.

Business Context

Imagine a retailer tracking its daily sales and profit for multiple store branches. Management wants to know which store performs best overall, how sales fluctuations affect profitability, and whether some stores are consistently more efficient than others.

The dataset store_sales.csv includes:

• Date – transaction date

• Store – store identifier

• Sales – daily sales revenue (GBP)

• Profit – daily profit (GBP)

Session Plan (2 Hours)

Time	Focus	Description
0 – 10 min	Introduction	Explain lab aims, libraries used, and dataset structure.
10 – 60 min	Tutor demonstration	Instructor leads guided examples using NumPy and pandas.
60 – 110 min	Student practice	Students complete structured analysis tasks.
110 – 120 min	Recap & reflection	Discuss findings and connect to next lab.

Hour 1 – Tutor-Led Demonstration

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Step 1: Import Libraries

Import and environment setup – show how analytical libraries are loaded in a notebook.

Step 2: Create a Simple NumPy Array

• Numeric arrays – illustrate how NumPy can store and analyse lists of numbers such as weekly sales.

Step 3: Compute Summary Statistics

• Summary measures – calculate and interpret averages, variation, and extremes in sales data.

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Step 4: Load the Dataset with pandas

• Loading a dataset – demonstrate reading the CSV file into pandas and viewing its structure.

• Inspection and profiling – examine column types, missing data, and descriptive statistics.

Step 5: Compute Store-Level Totals

• Aggregation – compute total sales and profit for each store and identify top performers.

Step 6: Sort and Display Top 5 Records

• Sorting and ranking – order the results to reveal the five highest-performing days or stores.

• Discussion – link these results to real-world decisions such as stock planning or staffing.

Hour 2 – Student Practice

1. Average performance – determine each store's average daily sales.
2. Relationship analysis – explore how strongly sales and profit are related.
3. Profit margin – calculate profit as a percentage of sales to compare efficiency.
4. High-value transactions – identify days with exceptionally large sales (above £12 000) and compare stores.

5. Visual summary – create a simple bar chart comparing total high-sales performance by store.
6. Reflection – consider which store appears most profitable and whether higher sales always imply higher profits.

Key Takeaways

• NumPy is the foundation for efficient numerical calculations.

• pandas provides intuitive tools for handling tabular business data.

• Aggregation and descriptive statistics quickly reveal performance patterns.

• Visual summaries help translate data into managerial insight.

• Python encourages reproducible and transparent analysis workflows.

Reflection Questions

• Which store demonstrates the most consistent performance?

• What might cause a store to have high sales but lower profits?

• How could this type of analysis support managerial decision-making?